

Generation Interconnection System Impact Study Report

For Queue Project Q13-0230

Desoto 230 kV

April 26, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0230. The Interconnection Customer proposes a 256.0 MW wind facility interconnecting at the Desoto 230 kV substation in the Indiana area. The request was received on April 21, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 178 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$54,753,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Baraga Vlg-Gray'S Flat 230.0 kV: 138.7% loading (519 MVA) following the outage of the Madeline-East Methuen Substation 230.0 kV Line; DFAX = 13.6%.

East Methuen-Madeline 230.0 kV: 122.3% loading (607 MVA) following the outage of the Baraga Vlg-Gray'S Flat 230.0 kV; DFAX = 14.3%.

Mason St-New Departure 230.0 kV: 115.6% loading (633 MVA) following the outage of the Trego-Beaver 500.0 kV; DFAX = 6.0%.

The Beaver 500/1kV TX is overloaded to 115.0% of its 1,356 MVA emergency rating for the loss of the Beaver-Eufaula 500.0 kV L. The project contributes 13.5% DFAX (34.6 MW) to this overload.

For the loss of Eufaula-Bever 500.0 kV, the Beaver Substation 230/1kV TX loads to 115.0% of its 1,356 MVA rating. Project DFAX: 13.5%.

Galion Generating Sta-Myrtle Point 230.0 kV: 114.5% loading (628 MVA) following the outage of the Trego-Beaver 500.0 kV L; DFAX = 6.5%.

For the loss of East ethuen-Henry County 500.0 kV line, the Murdock Substation-Sabanera 230.0 kV loads to 113.1% of its 562 MVA rating. Project DFAX: 41.3%.

The Eufaula 230 kV/1 kV Transformer Bank is overloaded to 112.2% of its 1,126 MVA emergency rating for the loss of the Imperial-Eufaula 500.0 kV. The project contributes 11.7% DFAX (30.0 MW) to this overload.

The Kampsville-Moreno Tap 230.0 kV is overloaded to 111.7% of its 570 MVA emergency rating for the loss of the Swan Valley Substation-Trego Substation 500.0 kV. The project contributes 8.5% DFAX (21.7 MW) to this overload.

Eastown Rd-Moreno Tap Substation 230.0 kV L: 111.3% loading (592 MVA) following the outage of the Swan Vly-Trego 500.0 kV; DFAX = 8.5%.

For the loss of Trego-Swan Vly 500.0 kV, the Zona Libre 6007-Kampsville 230.0 kV line loads to 111.2% of its 562 MVA rating. Project DFAX: 8.5%.

The Zona Libre 6007-George Neal S 230.0 kV line is overloaded to 111.0% of its 543 MVA emergency rating for the loss of the Swan Valley-Trego 500.0 kV line. The project contributes 8.5% DFAX (21.7 MW) to this overload.

The Myrtle Pt-George Neal South 230.0 kV is overloaded to 110.2% of its 528 MVA emergency rating for the loss of the Swan Vly-Trego 500.0 kV. The project contributes 8.5% DFAX (21.7 MW) to this overload.

Ft Humbug Substation 500/345kV Transformer: 109.1% loading (1,123 MVA) following the outage of the Orchard Rd-Fort Humbug 500.0 kV L; DFAX = 5.2%.

15Th Avenue S-Norwich 230.0 kV: 108.1% loading (507 MVA) following the outage of the Norwich-Omni-Hackberry 500.0 kV; DFAX = 6.6%.

Villa Park-Wolfe 230.0 kV line: 108.0% loading (703 MVA) following the outage of the Barrett-Wolfe 500.0 kV; DFAX = 6.0%.

The Washington County Cogeneration Facility-Villa Park 230.0 kV is overloaded to 107.4% of its 678 MVA emergency rating for the loss of the Barrett-Wolfe Substation 500.0 kV. The project contributes 6.0% DFAX (15.3 MW) to this overload.

The 15Th Avenue South-Lucent 230.0 kV line is overloaded to 107.3% of its 499 MVA emergency rating for the loss of the Norwich-Omni-Hackberry 500.0 kV. The project contributes 6.6% DFAX (16.8 MW) to this overload.

The Canal Rd-Newport Tie 230.0 kV L is overloaded to 105.3% of its 589 MVA emergency rating for the loss of the Trego Substation-Beaver 500.0 kV. The project contributes 10.0% DFAX (25.6 MW) to this overload.

The Eufaula 500/1kV Auto is overloaded to 105.2% of its 1,201 MVA emergency rating for the loss of the Imperial-Eufaula 500.0 kV Line. The project contributes 11.7% DFAX (30.0 MW) to this overload.

The Hilltop-Strawberry Belt 345.0 kV is overloaded to 105.1% of its 1,135 MVA emergency rating for the loss of the Hilltop Substation-Talenenergy Martins Crk Llc Williamsport 500.0 kV L. The project contributes 6.5% DFAX (16.7 MW) to this overload.

Carbon Center 500/345kV Auto: 105.0% loading (983 MVA) following the outage of the Blackrock-Shawville Station 345.0 kV L; DFAX = 7.3%.

Burris Pit-James De Young 230.0 kV L: 104.7% loading (557 MVA) following the outage of the Trego Substation-Swan Vly 500.0 kV; DFAX = 12.1%.

Hebo-Burris Pit 230.0 kV: 104.4% loading (542 MVA) following the outage of the Swan Vly-Trego 500.0 kV; DFAX = 12.1%.

Little Chute-Ft Humbug 345.0 kV line: 103.9% loading (1,440 MVA) following the outage of the Coronet Substation-Ft Humbug 500.0 kV; DFAX = 5.7%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Baraga Vlg-Gray'S Flat 230.0 kV	Rebuild line with higher-capacity conductor	\$31,836,000	6% (\$1,768,000)
East Methuen-Madeline 230.0 kV	Rebuild line with higher-capacity conductor	\$39,503,000	6% (\$2,181,000)
Mason St-New Departure 230.0 kV	Rebuild line with higher-capacity conductor	\$39,308,000	4% (\$1,698,000)
Beaver 500/1kV TX	Replace transformer with higher-rated unit	\$50,615,000	5% (\$2,407,000)
Beaver Substation 230/1kV TX	Replace transformer with higher-rated unit	\$10,883,000	5% (\$518,000)
Galion Generating Station-Myrtle Pt 230.0 kV	Rebuild line with higher-capacity conductor	\$29,875,000	5% (\$1,398,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Murdock Substation-Sabanera 230.0 kV	Rebuild line with higher-capacity conductor	\$29,510,000	23% (\$6,686,000)
Eufaula 230 kV/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$11,418,000	5% (\$534,000)
Kampsville-Moreno Tap 230.0 kV	Rebuild line with higher-capacity conductor	\$30,546,000	9% (\$2,750,000)
Eastown Rd Substation-Moreno Tap 230.0 kV	Rebuild line with higher-capacity conductor	\$62,995,000	9% (\$5,671,000)
Zona Libre 6007-Kampsville 230.0 kV	Rebuild line with higher-capacity conductor	\$28,829,000	9% (\$2,595,000)
George Neal S-Zona Libre 6007 230.0 kV	Rebuild line with higher-capacity conductor	\$27,451,000	9% (\$2,471,000)
Myrtle Pt-George Neal South 230.0 kV	Rebuild line with higher-capacity conductor	\$14,524,000	9% (\$1,308,000)
Ft Humbug 500/345 kV TX	Replace transformer with higher-rated unit	\$25,225,000	44% (\$11,230,000)
15Th Avenue S-Norwich 230.0 kV	Reconductor line	\$4,599,000	7% (\$338,000)
Wolfe-Villa Park 230.0 kV line	Reconductor line	\$6,684,000	10% (\$672,000)
Washington County Cogeneration Facility-Villa Park 230.0 kV	Reconductor line	\$11,558,000	10% (\$1,162,000)
Lucent-15Th Avenue South 230.0 kV	Reconductor line	\$3,614,000	7% (\$266,000)
Canal Rd-Newport Tie 230.0 kV	Reconductor line	\$8,535,000	5% (\$397,000)
Eufaula 500-1 kV Transformer Bank	Replace transformer with higher-rated unit	\$30,336,000	5% (\$1,418,000)
Strawberry Belt-Hilltop 345.0 kV Line	Reconductor line	\$24,652,000	5% (\$1,177,000)
Carbon Center 500/345kV Auto	Replace transformer with higher-rated unit	\$36,220,000	5% (\$1,721,000)
Burris Pit-James De Young 230.0 kV	Reconductor line	\$16,978,000	6% (\$1,046,000)
Hebo-Burris Pit 230.0 kV	Reconductor line	\$9,489,000	6% (\$585,000)
Little Chute Substation-Ft Humbug 345.0 kV	Reconductor line	\$46,342,000	6% (\$2,756,000)
	Total allocated to project		\$54,753,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.